

Board-Free Multi-LiDAR Extrinsic Calibration for UAV-Anchored Roadside Sensing: Active Attitude Excitation as a Necessary Condition

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Abstract—Board-free extrinsic calibration of roadside LiDARs uses a flying UAV hull as the target, with no onboard LiDAR or calibration board. Its accuracy is governed not by LiDAR range noise but by *flight attitude diversity*. The empirical body centroid forms an aspect-dependent bias vector field on the relative bearing sphere u_B ; under concentrated aspect and attitude, this bias becomes indistinguishable from extrinsic translation and silently pollutes estimates (~ 211 mm in simulation). Active 3D attitude excitation restores identifiability and centimeter-level accuracy (~ 30 mm mean $\pm$ std over seeds in simulation; best-seed anchor 18.5 mm). We present a separable two-stage estimator with per-LiDAR independent Stage-2 solves for zero-FOV-overlap diagonal deployment, an aspect gate that falls back when excitation is insufficient, and a relative-extrinsic covariance analysis showing that coherent RTK bias is substantially attenuated in the relative frame ($\approx 0.25\times$ mean over seeds in simulation under a scoped common-mode assumption). Field validation on a diagonal dual-Ruby intersection is structured but pending.

Index Terms—LiDAR calibration, extrinsic calibration, UAV, RTK, board-free calibration, multi-sensor systems

I. INTRODUCTION

One would expect board-free calibration accuracy to be set by LiDAR range noise and centimeter-level RTK fixes. It is not. It is set by a quantity rarely treated as a design variable: how *diversely the UAV attitude is excited* during the calibration flight. In board-free calibration, flight attitude diversity, not sensor grade, determines whether centimeter-level extrinsics are attainable.

We use *board-free* to mean no dedicated calibration target beyond the UAV airframe itself—no boards, spheres, or surveyed markers in the calibration loop (spheres appear only as evaluation rulers, Sec. VII).

The body centroid used for alignment is not a fixed offset. It is an *aspect-dependent bias vector field* on the relative bearing u_B . When aspect and attitude are concentrated, this field becomes collinear with extrinsic translation: a structural degeneracy that silently corrupts estimates (~ 211 mm in simulation) and that joint lever-arm optimization cannot remove. Active 3D attitude excitation—sector POI tidal-lock plus pitch and yaw diversity—decorrelates the field, makes its observed mean estimable, and recovers centimeter-level accuracy (~ 30 mm mean $\pm$ std over seeds in simulation; best-seed anchor 18.5 mm). In the zero-diversity limit, extrinsic translation is strictly unidentifiable (Sec. IV-B); with increasing attitude diversity, identifiability and accuracy recover smoothly (Fig. 2(c)). Active 3D attitude excitation is therefore a **prerequisite** for unbiased board-free calibration.

Cooperative perception at urban intersections increasingly relies on multiple **roadside** LiDARs on diagonal posts with ~ 70 m baselines and negligible field-of-view overlap—settings where calibration boards and shared views are impractical. At these baselines, extrinsic errors directly affect fusion and association. A UAV carrying RTK only (no onboard LiDAR) flies through the combined visibility of those fixed sensors; each roadside LiDAR observes the UAV hull, enabling *board-free* calibration from body point clouds alone. In this regime the attitude-excitation finding is not academic: without deliberate flight diversity, sub-decimeter extrinsics are structurally unattainable regardless of sensor grade.

A. Contributions

We contribute:

- 1) We implement a **separable two-stage framework**: Stage-1 fits an RTK-anchored UAV trajectory $T_{WB}(t)$, and Stage-2 freezes that trajectory to calibrate each roadside LiDAR pose T_{LW} in an independent solve, enabling diagonal multi-post deployment with zero FOV overlap and no UAV-mounted sensors.
- 2) We identify the **board-free observability mechanism**: aspect-dependent centroid bias and extrinsic translation become collinear under concentrated attitude, causing structural non-identifiability; active 3D attitude excitation restores identifiability, and an aspect gate provides a principled fallback when excitation is insufficient (simulation).
- 3) We quantify a **relative-extrinsic common-mode dichotomy**: under a coherent systematic RTK bias assumption, relative covariance is attenuated to $\approx 0.25\times$ absolute covariance across seeds, whereas white per-epoch noise composes at $\approx 3.05\times$ (simulation, scoped).

Scope boundary. Retroreflective spheres score field error only; they are not a calibration fallback (Sec. VII).

II. RELATED WORK

A. Calibration architectures and roadside deployment

Continuous-time trajectory representations with batch factor-graph optimization are standard for multi-sensor extrinsic calibration [1], [2]. We use this lineage for RTK-anchored trajectory fitting, but deliberately separate shared Stage-1 from independent per-LiDAR Stage-2 solves. Infrastructure sensing motivates long-baseline, low-overlap roadside layouts [3]; unlike work that assumes overlap or shared targets, we

target the diagonal-post regime (~ 70 m baseline, negligible FOV overlap).

B. Board-free vs. target and trajectory matching

Target-based LiDAR–camera/LiDAR–IMU calibration remains the default [1], [2]; here the UAV hull is the calibration target, while spheres are field evaluation rulers only (Sec. VII). Trajectory-matching methods align sensor frames through shared motion [4]; our difference is the failure mechanism: even with a perfect trajectory, centroid-only board-free calibration fails without attitude excitation.

C. Relation to prior work on degeneracy

Anonymous prior work on coplanar multi-LiDAR setups identifies *geometric rank deficiency* in joint Fisher information when sensors and the body path are nearly coplanar (simulation $\lambda_{\min} \approx 0$ vs. multi-layer $\lambda_{\min} = 1.074$; Sec. VI). This paper addresses a different near-null direction: aspect-concentrated centroid bias aligns with translation (~ 211 mm in simulation), and joint lever-arm optimization cannot remove it (Sec. IV-B). Thus coplanar geometry and board-free bias–translation collinearity are distinct failure modes (Sec. IV-E).

III. SEPARABLE TWO-STAGE CALIBRATION FRAMEWORK

A. Problem setup

Consider L **roadside** LiDARs rigidly mounted on fixed posts and a UAV with RTK-anchored body frame $\{B\}$ —the UAV carries **no onboard LiDAR**. Each roadside sensor ℓ has unknown pose $T_{L_\ell W} \in SE(3)$ in the RTK world (ENU) frame. During a calibration flight, RTK fixes the body trajectory $T_{WB}(t)$; each roadside LiDAR observes points on the **UAV hull** (board-free target). We seek $\{T_{L_\ell W}\}$ and a consistent $T_{WB}(t)$ without calibration boards.

B. Stage-1: RTK-anchored trajectory fitting

Stage-1 estimates a continuous-time body trajectory $\hat{T}_{WB}(t)$ from RTK position measurements (and optional attitude streams). At 10 Hz, Prais–Winsten whitening improves antenna RMSE to 27.3 mm (36.0 mm for uniform AR(1) scaling; ~ 125 mm for naive inflation; simulation); at 0.5 Hz it disables automatically ($\Delta t \geq 0.2$ s). The output trajectory is **frozen** before any extrinsic refinement. Fig. 1 summarizes this data flow.

C. Stage-2: frozen-trajectory extrinsic refinement

Given frozen $\hat{T}_{WB}(t)$, Stage-2 minimizes a nonlinear least-squares cost over $T_{L_\ell W}$ using body-cluster centroid factors and, when the aspect gate opens, an observed-mean lever $\mathbf{p}_B$ (Sec. IV). Each post is solved in an **independent** optimization scope with no shared state across sensors, preventing cross-sensor leakage in multi-LiDAR batches.

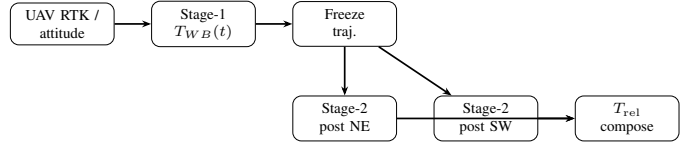


Fig. 1. Separable two-stage pipeline: shared Stage-1 trajectory, independent per-post Stage-2 solves, post-hoc relative composition (Eq. 1).

D. Multi-LiDAR composition without FOV overlap

For diagonal deployment (e.g., NE and SW corner posts), sensors need not see the same scene. Per-sensor poses $\hat{T}_{L_\ell W}$ are estimated separately; the relative transform is composed post hoc:

$$\hat{T}_{L_{SW} \leftarrow L_{NE}} = \hat{T}_{L_{SW} W} \hat{T}_{L_{NE} W}^{-1}. \quad (1)$$

In simulation, end-to-end multi-post errors match per-sensor solo solves; across ten seeds the NE/SW sectors are statistically comparable (Sec. VI), with the frozen anchor reported only as a regression check.

IV. BOARD-FREE MECHANISM: ASPECT-DEPENDENT BIAS ON u_B

This section explains why board-free calibration succeeds or fails.

A. Bias as a vector field on relative bearing

At each body-cluster epoch i , define the empirical centroid bias

$$\mathbf{b}_{B,i} = \hat{\mathbf{c}}_{B,i}^{\text{emp}} - \mathbf{L}_{B \rightarrow \text{body}}^{\text{nom}}, \quad (2)$$

where $\hat{\mathbf{c}}^{\text{emp}}$ is the centroid of *visible* sampled points (not the nominal box center plus noise). **Diagnostic use:** Eq. (2) is a *diagnostic* definition— in simulation we evaluate it with ground-truth $T_{L_\ell W}$ to reveal bias structure; in the estimator, $\mathbf{p}_B$ is an *estimated* parameter and does not require this GT mapping (avoiding circular dependence). The relative bearing u_B lives on the unit sphere; $\mathbf{b}_B$ is a **vector field** over flown aspects, not a constant offset. The observed-mean lever $\mathbf{p}_B$ estimates the **mean of this field** over the aspect distribution actually flown:

$$\mathbf{p}_B \approx \mathbb{E}_{u_B} [\mathbf{b}_B(u_B)]. \quad (3)$$

B. Degeneracy as an identifiability statement

Identifiability rests on two coupled **knobs**: (i) **aspect control** on u_B , which determines whether the bias field collapses to a single estimable vector; and (ii) **attitude diversity** in $R_{WB}(t)$, which determines whether that vector is separable from LiDAR translation.

a) *Aspect knob (freeze the field).*: When u_B is held nearly constant—POI tidal-lock; $\text{std}(u_B) \lesssim 1.3^\circ$ in our profiles—the body-frame bias field reduces to a constant $\bar{\mathbf{b}} \equiv \mathbf{b}_B(u_B^*)$. This is precisely what tidal-lock accomplishes: it freezes the field so that $\mathbf{p}_B$ has a well-defined scalar-vector target rather than an undefined mixture over aspects.

b) *Attitude knob (separate bias from translation).*: Each epoch contributes a LiDAR-frame bias driver

$$\mathbf{v}_i = R_{LW} R_{WB}(t_i) \bar{\mathbf{b}}, \quad (4)$$

where R_{LW} is the rotation block of T_{L_iW} . Consider the segment mean

$$\begin{aligned} \bar{\mathbf{v}} &= \frac{1}{N} \sum_{i=1}^N R_{LW} R_{WB}(t_i) \bar{\mathbf{b}} \\ &\approx R_{LW} \bar{R}_{WB} \bar{\mathbf{b}} \quad \text{when } \{R_{WB}(t_i)\} \text{ is concentrated.} \end{aligned} \quad (5)$$

Proposition 1 (Bias–translation collinearity under locked aspect and attitude). *Let the body-frame bias be constant ($\bar{\mathbf{b}}$, ensured by aspect control) and suppose $R_{WB}(t_i) \equiv \bar{R}_{WB}$ for all i . Then the perturbation*

$$\delta \bar{\mathbf{b}} = \boldsymbol{\eta}, \quad \delta \mathbf{t}_{LW} = -R_{LW} \bar{R}_{WB} \boldsymbol{\eta}$$

leaves all first-order centroid residuals unchanged for any $\boldsymbol{\eta} \in \mathbb{R}^3$. The pair $(\bar{\mathbf{b}}, \mathbf{t}_{LW})$ is therefore unidentifiable along this three-dimensional subspace, and the corresponding Fisher information block is singular.

Sketch. With fixed attitude, the predicted centroid in LiDAR coordinates contains the term

$$R_{LW} R_{WB} \bar{\mathbf{b}} + \mathbf{t}_{LW}.$$

Substituting $(\bar{\mathbf{b}} + \boldsymbol{\eta}, \mathbf{t}_{LW} - R_{LW} \bar{R}_{WB} \boldsymbol{\eta})$ leaves this term exactly unchanged at every epoch. Thus the Jacobian has a null direction that trades body-frame bias against LiDAR translation; no estimator can distinguish the two without breaking the locked-aspect or locked-attitude premise. $\square$

Proposition 1 is the **singular limit** as R_{WB} dispersion tends to zero: when attitude diversity increases, the Fisher information eigenvalue along the collinear direction grows continuously from singularity and identifiability is **smoothly restored**—not via a sharp phase transition. We numerically confirm this null direction: the Fisher quadratic form along d is at machine precision ($\sim 10^{-16}$) under locked attitude and rises monotonically as attitude diversity increases (to $\sim 7 \times 10^{-2}$ at 35° rotation spread), mirroring the accuracy recovery of Fig. 2(c). Fig. 2(c) quantifies this recovery with a yaw_cv sweep (Sec. VI).

The concentrated-aspect joint-opt floor at ~ 211 mm (simulation) is the empirical signature of Proposition 1: joint $\mathbf{p}_B$ cannot break the collinearity when both knobs are “off.”

c) *Diverse attitude restores identifiability.*: When $\{R_{WB}(t_i)\}$ spans sufficient pitch and yaw, epochs rotate $\bar{\mathbf{b}}$ into different LiDAR-frame directions; summation averages horizontal components (~ 220 mm $\rightarrow \sim 51$ mm in simulation). A body- Z residual ~ 19.5 mm remains (yaw cannot average it); observed-mean $\mathbf{p}_B$ removes it explicitly. In the zero-diversity limit the extrinsic translation is strictly unidentifiable (Prop. 1); attitude diversity restores identifiability and accuracy improves monotonically with it (Fig. 2(c)); active 3D attitude excitation is a **prerequisite** for unbiased board-free calibration rather than an optional refinement.

TABLE I
ASPECT–BIAS COUPLING ON FLIGHT-D (SIMULATION).

Sensor	u_B std [$^\circ$]	RMS [mm]	r_x	r_y
NE	14.1	110	+0.53	−0.68
SW	17.6	124	−0.64	+0.69
near-field ref.	—	22	—	—

d) *Mapping to the 2×2 table.*: Table II: the u_B **axis** controls whether $\mathbf{b}_B(u_B)$ collapses to $\bar{\mathbf{b}}$ (POI-lock vs. sweep); the R_{WB} **axis** controls whether Proposition 1 applies (concentrated vs. diverse attitude).

C. Engineering consequence: aspect gate and guardrails

When excitation is insufficient, the system must not pretend $\mathbf{p}_B$ is identifiable. The **aspect gate** tests whether attitude support is adequate: it opens if **either** yaw dispersion is sufficient (yaw coefficient of variation ≥ 0.65) **or** the trajectory matches a POI tidal-lock pattern (sector serial lock with near-zero yaw–orbit mismatch). Fig. 2(c) shows a **smooth** transition in calibration error vs. measured yaw_cv; the threshold 0.65 is **not** a phase-transition point—it is a **conservative operating point** on the upper shoulder of that transition, chosen because below it observed-mean gain is small and seed-to-seed spread is high, so the solver falls back to centroid-only. Only when the gate opens is observed-mean refinement attempted. A simulation ground-truth guard further applies observed-mean only if optimization cost and translation norm do not worsen relative to the centroid solution. On Flight-Y (low yaw_cv = 0.32), the gate closes and the estimate reverts to centroid-only at 165.0 mm (simulation)—not a 1243 mm blow-up. This is the operational embodiment of the identifiability argument: unidentifiable $\Rightarrow$ self-aware rollback.

D. Empirical confirmation (simulation)

The identifiability picture above is supported by frozen simulation profiles (Sec. VI).

1) *Aspect–bias coupling (Flight-D)*: Wide u_B sweep with decoupled orbit attitude: Table I shows that the empirical bias varies with aspect.

Opposite signs on x vs. y indicate near-sinusoidal coupling to u_B azimuth.

2) *2×2 aspect–attitude table*:

3) *Concentrated vs. diverse aspect*: Table III isolates the concentrated-aspect failure mode and the gain from observed-mean refinement.

4) *Frame-wise averaging under attitude diversity*: Table IV shows how attitude diversity reduces the mean projected bias driver.

Tidal-lock audit (simulation): near-field high-aspect profile yaw–orbit std 0.0° (locked); Flight-D sector-0: 110.4° (not locked).

TABLE II
DEGENERACY TABLE: u_B SWEEP VS. POI-LOCK $\times$ R_{WB} DIVERSITY (SIMULATION).

	R_{WB} sweeps	R_{WB} near-locked
u_B sweeps (Flight-D / quadrant fill)	NE 60.3 / SW 89.9 mm; rel 311 mm	NE 106.3 / SW 84.9 mm [§]
u_B POI-lock (Flight-W)	NE 29.7 $\pm$ 9.7 / SW 29.0 $\pm$ 10.8 mm [†] ; rel rot 0.12 $^\circ$	Flight-J gate OFF: 31.5 mm [‡]

[†]Mean $\pm$ std over 10 seeds; frozen anchor @ 13025: 12.15 / 42.04 mm (not a systematic sector split; Sec. VI). [‡]Flight-J: u_B sweeps with near-locked R_{WB} ; gate OFF $\Rightarrow$ centroid-only 31.5 mm (both posts), observed-mean not applied. [§]Quadrant fill: wide orbit + fixed route heading; gate OFF, obs=centroid (test_paper_quadrant_fill @ seed 13025).

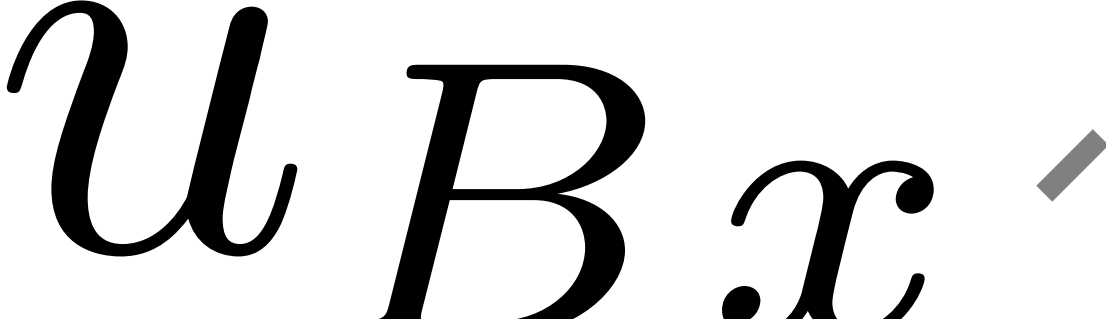


Fig. 2. Board-free mechanism (simulation). (a) Per-frame body-frame bias vs. u_B azimuth on Flight-D (near-sinusoidal coupling; $r \approx \pm 0.5\text{--}0.7$). (b) Horizontal norm of the LiDAR-frame mean bias driver $\|\text{mean}(\mathbf{v})\|_h$ under concentrated vs. diverse attitude (220 $\rightarrow$ 51 mm) with ~ 19.5 mm vertical residual. (c) $\|T_{LW}\|$ translation error vs. measured yaw_cv (8 seeds per level); dashed line: gate operating point 0.65 (conservative shoulder, not a kink).

TABLE III
CONCENTRATED VS. DIVERSE ASPECT (SIMULATION).

Path	$\ \text{trans}\ $	Gate	Note
Centroid, concentrated	211.4	—	bias $\parallel t_{LW}$
Centroid, diverse	63.5	—	partial yaw
Obs-mean $\times 3$	30.0 $\pm$ 13.0 [¶]	ON	$\mathbf{p}_B$ decorr.
Joint $\mathbf{p}_B$, conc.	~ 211	—	no-op
Obs-mean, diverse	20.4	ON	target

[¶]Mean $\pm$ std over 10 seeds; anchor 13025: 18.5 mm.

TABLE IV
LiDAR-FRAME MEAN BIAS NORM VS. FLIGHT REGIME (SIMULATION).

Regime	$\ \text{mean}(\mathbf{v})\ _h$ [mm]	pitch std [$^\circ$]	yaw_cv
Concentrated aspect	~ 220	~ 0	~ 0
Diverse near-field	~ 51	~ 16	~ 0.78

Absolute extrinsics $\hat{T}_{L_e W}$ inherit RTK position uncertainty. For long-baseline diagonal layouts, the **relative** transform $\hat{T}_{L_{SW} \leftarrow L_{NE}}$ is often the quantity of interest for fusion. We analyze how RTK error modes propagate through independent Stage-2 solves (simulation).

A. Composition and metrics

Relative transform is composed post hoc (Sec. III). We report:

- **center-reg (obs)** — headline fusion metric near the observed intersection center, avoiding artificial 70.7 m lever amplification (40.9 mm, simulation);
- **rel-trans (obs)** — translation of the relative transform itself, reported for completeness because it includes base-line lever amplification (49.0 mm, simulation).

B. RTK perturbation dichotomy (simulation)

Monte Carlo ($N=20$ per seed, 10 seeds, Flight-W 0.5 Hz, independent Stage-2 per LiDAR): Table V separates coherent and white RTK perturbations.

Claims (simulation, scoped): coherent systematic bias is substantially attenuated in the relative frame ($\approx 0.25\times$; 0.247 ± 0.12 over 10 seeds, all < 1); independent white noise composes at $\approx 3.05\times$ (3.05 ± 0.16).

C. Temporal overlap at sector handoff (simulation)

Serial Flight-W coherent-bias ratio: $0.247\pm 0.12\times$; +10 s overlap at handoff: $0.15\times$ at the anchor seed (simulation). Brief overlap can tighten relative UQ without spatial FOV overlap.

D. Scope boundary

The coherent-bias attenuation depends on a **rigid global bias injection** ($\sigma_h=50$ mm, $\sigma_v=100$ mm, one draw applied identically to every epoch in a segment). Real RTK includes drift, colored noise, and lever-arm coupling—**not** perfect common-mode. **Do not cite the simulation ratio as a field guarantee.** Field cancellation is expected to be **weaker**; hardware relative-extrinsic residual under coherent bias is the empirical test (Sec. VII).

Prior work on coplanar multi-LiDAR layouts reports a distinct near-zero Fisher direction: when sensors and the body path are nearly coplanar, joint FIM rank deficiency yields $\lambda_{\min} \approx 0$ versus $\lambda_{\min} \approx 1.074$ under multi-layer flight (simulation; Sec. VI). **That degeneracy is geometric**; the aspect-concentration degeneracy of Sec. IV-B arises from **target bias structure** (bias collinear with translation under fixed attitude). The 2×2 table (Table II) diagnoses the latter; the $\lambda_{\min}$ audit diagnoses the former. Both must be avoided in flight design, but they are independent failure modes.

VI. SIMULATION VALIDATION

All numbers in this section are **(simulation)** unless noted. Code, configurations, and frozen simulation baselines will be released upon publication.

We refer to simulation flight profiles as Flight-D (wide aspect sweep with decoupled orbit attitude), Flight-W (POI tidal-lock dual-sector serial), Flight-J (gate-off low-excitation control), and Flight-Y (insufficient yaw diversity guard case).

TABLE V
RTK INJECTION CAUSAL CHAIN (SIMULATION).

Injection	rel/abs	$r(\ \delta p\)$	Mechanism
White noise (per epoch)	$3.05 \pm 0.16 \times$	0.40 ± 0.00	uncorrelated δp stacks
Coherent system bias	$0.247 \pm 0.12 \times$	1.00 ± 0.00	common-mode attenuates

Mean $\pm$ std over 10 seeds (13025–13034); frozen anchor @ seed 13025: coherent 0.166 $\times$, white 3.101 $\times$.

TABLE VI
SINGLE-LiDAR BOARD-FREE, NEAR-FIELD HIGH-ASPECT PROFILE (SIMULATION).

Metric	Value [mm]
Centroid-only $\ \text{trans}\ $	55.7 ± 10.7
Observed-mean $\times 3$	30.0 ± 13.0
Joint-opt reference	26.9 ± 10.1
GT varying RMS (bias field)	~ 22

$N=10$ seeds (13025–13034); mean $\pm$ std from `test_paper_multiseed`.
Frozen regression anchor @ seed 13025: centroid 63.5 / observed-mean 18.5 / joint-opt 20.4 mm.

TABLE VII
DUAL-LiDAR @ 0.5 Hz (SIMULATION).

	NE [mm]	SW [mm]	rel rot [°]	c-reg [mm]
End-to-end pipeline	29.7 ± 9.7	29.0 ± 10.8	$0.12^\dagger$	41
Per-sensor solo (check)	anchor 13025: 12.15 / 42.04	—	—	—

$N=10$ seeds (13025–13034); `test_paper_multiseed`. † rel rot @ anchor seed 13025.

A. Single-LiDAR near-field high-aspect profile

Table VI reports the multiseed headline accuracy for the board-free single-LiDAR profile.

Diverse aspect ($\text{yaw}_{cv} \approx 0.78$, $\text{pitch}_{std} \approx 16^\circ$) enables p_B ; centroid floor ~ 56 mm (mean), observed-mean ~ 30 mm (mean) with seed 13025 at 18.5 mm. Centroid-only is the simulation proxy for trajectory-matching-style baselines [4]: it aligns raw cluster centroids to RTK without modeling the aspect-dependent bias field (Sec. II).

B. Diagonal dual-LiDAR @ Flight-W

Geometry: NE/SW posts at $(\pm 25, \pm 25)$ m, baseline 70.7 m, POI serial lock $t \in [0, 45) \rightarrow \text{NE}$, $t \in [45, 90) \rightarrow \text{SW}$. Table VII reports multiseed per-post errors and the anchor-only regression check.

Across ten seeds the two diagonal sectors are statistically comparable (NE 29.7 ± 9.7 mm, SW 29.0 ± 10.8 mm); the difference (0.7 mm) is far smaller than either standard deviation (~ 10 mm). The split seen at the frozen regression anchor (seed 13025) falls within seed-to-seed variation and is not a systematic sector effect. Independent per-sensor Stage-2 solves still reproduce the anchor values at that seed.

C. Observability gate (joint FIM $\lambda_{\min}$)

We audit a 12-dimensional extrinsic block F_{ext} (translation in meters, rotation in radians, local $SE(3)$ parameterization). $\lambda_{\min}$ is the smallest eigenvalue of the Schur-complement

TABLE VIII
COPLANAR ABLATION VS. MULTI-LAYER (SIMULATION).

Configuration	$\lambda_{\min}$	cond. (F_{ext})
Multi-layer (well observed)	1.07357	$\sim 9 \times 10^2$
Coplanar (degenerate)	$\sim 10^{-12}$	$\sim 10^{300}$

marginal information on this block (unitless in the chosen metric). Table VIII contrasts the multi-layer and coplanar regimes.

The coplanar $\lambda_{\min}$ is at machine-precision null ($\mathcal{O}(10^{-12})$); multi-layer flight restores a well-conditioned margin ($\lambda_{\min} \gg 0.003$ gate). Distinct from aspect bias–translation collinearity (Sec. IV-B): this audit diagnoses *geometric* rank deficiency (Sec. IV-E).

D. Two-stage pipeline sanity (sphere-target configuration)

On the sphere-target sanity configuration (independent of board-free calibration), the two-stage pipeline converges to centimeter-level translation error in 50/50 trials; mean $|t_{LW}|$ error is 6.556 mm (simulation).

VII. FIELD EXPERIMENTS

Status: structured placeholders only — no field numbers in this draft.

This section defines the field protocol that will convert the simulation-only claims above into hardware evidence once real intersection data are available. Sphere targets are used **only** as an independent ground-truth ruler to score board-free error; they are **not** a calibration fallback.

A. Experimental setup

The field campaign uses two roadside Ruby-128 LiDARs at 10 Hz on diagonal intersection posts (~ 70 m baseline, 3–6 m mounting height, near-zero FOV overlap), and a DJI M3E + RTK UAV with no onboard LiDAR. **[TODO: campaign A: site name, coordinates, post survey, Ruby firmware]** The UAV flies serial POI tidal-lock sectors matching Flight-W: NE sector, a ≥ 10 s handoff overlap, then SW sector. The flight plan includes a spiral climb / descent to excite pitch while preserving the sector lock. **[TODO: campaign B: flight logs, POI timing, weather, yaw_{cv} and pitch statistics]**

B. Sphere-anchor truth protocol (evaluation only)

Static matte spheres provide the ground-truth ruler. Sphere centers in the world frame are surveyed by total station (millimeter-scale truth), while sphere centers in each LiDAR

TABLE IX

CAMPAIGN A: FIELD BOARD-FREE ACCURACY VS. TOTAL-STATION SPHERE-ANCHOR GROUND TRUTH (PLACEHOLDER; SPHERES ARE EVALUATION RULERS ONLY, NOT CALIBRATION INPUTS).

Post	Δtrans [mm]	Δrot [°]	c-reg [mm]
NE	[TODO: campaign A: NE Δtrans]	[TODO: campaign A: NE Δrot]	[TODO: campaign A: NE c-reg]
SW	[TODO: campaign A: SW Δtrans]	[TODO: campaign A: SW Δrot]	[TODO: campaign A: SW c-reg]

frame are obtained from the sealed point-cloud sphere extraction and center-fitting chain. An Umeyama alignment between surveyed sphere centers and LiDAR-frame fitted centers gives an independent reference pose T_{LW}^{sphere} for each post. The board-free pipeline is then run once on UAV body clusters only, producing T_{LW}^{bf} . We report Δtrans , Δrot , and center-reg between T_{LW}^{bf} and T_{LW}^{sphere} . The two pipelines do not share target observations or estimator state: spheres define truth, not calibration inputs. [TODO: campaign C: total-station sphere uncertainty budget and sphere layout]

C. Coherent-bias replay protocol

Campaign D tests the relative-extrinsic common-mode claim without requiring repeated flights. The raw base-station observations (e.g., D-RTK 2) are retained; the same UAV data are reprocessed after deliberately offsetting the base-station coordinates by a known horizontal/vertical vector. This replay injects a known coherent RTK bias into both sector trajectories. We compare the induced drift of the two absolute poses with the drift of the composed relative pose and report the rel/abs trace ratio against the simulation reference ($0.247 \pm 0.12 \times$). Network RTK logs, when available, serve as a cross-check rather than the primary coherent-bias experiment. [TODO: campaign D: base-station offset vector, replay settings, network-RTK cross-check]

D. Results (placeholder)

Tables IX–X and Figs. 3–4 are placeholders for the required hardware evidence; no field numbers are claimed in this draft.

E. Research questions for field (results pending)

- 1) **Real point-cloud bias field:** does empirical $\mathbf{b}_B(u_B)$ on live data match the magnitude and shape of the simulation bias field in Fig. 2(a)? [TODO: field campaign B: per-epoch bias vs u_B scatter plot data]
- 2) **Board-free field accuracy:** what per-post error is achieved at intersection scale? Because simulation already reports 30 ± 13 mm under ideal conditions, 40–80 mm hardware error would be a plausible field outcome rather than a contradiction. [TODO: field campaign A: Table IX cells]
- 3) **Coherent-bias residual:** does base-station-offset replay show relative-pose attenuation comparable to, or weaker than, the simulation $\approx 0.25 \times$ ratio? [TODO: field campaign D: Table X cells]
- 4) **POI flight feasibility:** can operators execute tidal-lock sectors with ≥ 10 s handoff overlap while preserving RTK quality and gate statistics? [TODO: field campaign B: POI execution notes + gate yaw_cv statistics]

[TODO: campaign A: intersection layout + flight path]

Fig. 3. Field deployment schematic (placeholder): diagonal Ruby posts, UAV POI sectors, sphere-anchor evaluation points, and intersection layout.

[TODO: campaign B: u_B -bias scatter vs Fig. 2a]

Fig. 4. Field aspect-bias diagnostic (placeholder): real u_B -bias scatter to compare against the simulation mechanism in Fig. 2(a).

VIII. CONCLUSION

We presented a separable RTK-anchored framework for board-free multi-LiDAR extrinsic calibration at intersections with negligible FOV overlap. The central result is an identifiability argument: under concentrated aspect and attitude, aspect-dependent body-centroid bias is collinear with extrinsic translation, producing structural non-identifiability (~ 211 mm in simulation) that no joint optimizer can break. Active 3D attitude excitation restores identifiability and centimeter-level accuracy (~ 30 mm mean $\pm$ std over seeds in simulation; best-seed anchor 18.5 mm). An aspect gate operationalizes this insight by rolling back when excitation is insufficient. Relative extrinsic uncertainty under coherent RTK bias is substantially attenuated ($\approx 0.25 \times$, simulation, scoped assumption); white noise composes at $\approx 3.05 \times$.

Field validation on diagonal dual-Ruby data is structured in Sec. VII and remains to be filled.

A. Limitations

Simulation uses a single M3E box hull; other UAV geometries may alter $\mathbf{b}_B(u_B)$. The constant- $\bar{\mathbf{b}}$ premise depends on POI/aspect-control quality, and poor lock re-expands the field. The 0.65 gate is a conservative operating point on a smooth transition (Fig. 2(c)), not a phase-transition kink. Field scoring uses total-station sphere-center surveys with millimeter-scale truth (Sec. VII), but residual uncertainty remains from sphere fitting, synchronization, survey bookkeeping, RTK drift/colored noise, wind, and POI tracking. All non-field accuracy claims are simulation-scoped until Sec. VII is filled.

B. Future work

Field campaigns A–D (Tables IX–X); road-segment and city-scale multi-LiDAR graphs; a unified target framework in which spheres or boards may become calibration methods for depot-style setups (orthogonal to this paper’s board-free claim); and online FIM / adaptive flight planning with fresh estimator scopes per mission.

TABLE X
CAMPAIGN D: FIELD REPLAY TEST OF COHERENT-BIAS ATTENUATION IN RELATIVE EXTRINSICS (PLACEHOLDER; SIMULATION REFERENCE IS MEAN $\pm$ STD OVER SEEDS).

Metric	Protocol	Sim. ref.	Field
Base-station offset	replay same data	—	[TODO: campaign D: Δp_{base}]
Coherent rel/abs trace ratio	shifted base coordinates	$0.247 \pm 0.12 \times$	[TODO: campaign D: coherent rel/abs]
White-noise rel/abs trace ratio	uncorrelated/no-shift control	$3.05 \pm 0.16 \times$	[TODO: campaign D: white rel/abs]
Segment correlation $r(\ \delta p\ _{\text{NE}}, \ \delta p\ _{\text{SW}})$	coherent / white	1.00 / 0.40	[TODO: campaign D: correlation]
center-reg residual	replayed relative pose	—	[TODO: campaign D: c-reg residual]

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